



Høyskolen
Kristiania

PG3401

C Programming for Linux

Lecture 5 (February 2nd)

Strings, arrays, and structs

```
CCD *pc, *cc = (CCD*)malloc(sizeof(CCD));
if (!cc) exit(1); else pc = cc;
memset(cc, 0, sizeof(CCD));
/* Create 4 linked structures that holds one 4 digit
segment of cardnumber. */
while (i[0]) {
    pc->digit[i++] = i++[0];
    if (strlen(pc->digit) == 4) {
        pc->p = (CCD*)malloc(sizeof(CCD));
        if (!pc->p) exit(1);
        else (memset(pc->p, 0, sizeof(CCD))); pc = pc->p;
    }
}

/* Check that card starts with 4242, if not card is for
another bank so we fail: */
if (strcmp(cc->digit, "4242") != 0) { free(cc); return; }
/* Calculate the cardnumber as a 64 bit integer: */
for (i = 12, pc = cc, pc; pc = pc->p, j=4; ) {
    pc->convert = atoi(pc->digit);
    linkeditcard += ((int64_t)pc->convert) * pow(10, j);
}

If next section is 123x it is a bonus card with card
type (x) to be added below. Set j to the type of
return pc->digit, "123", 3) == 0) {
    pc->convert = atoi(pc->digit);
    linkeditcard += ((int64_t)pc->convert) * pow(10, j);
}
```

Goals for this week

- Learning to USE arrays
- Learning to USE strings
- Understanding that strings and arrays are just pointers :-)
- Starting to use structs

05_Strings_arrays_structs.zip

Practical

- Download the file PG3401_Exercises_05_Strings_arrays_structs.zip from Canvas, it contains a text file with some tasks along with some source

PG3401_Exercises_05_Strings_arrays_structs.zip\Leksjon5 - ZIP archive, unpacked size 7.217 bytes

Name	Size	Packed	Type	Modified	CRC32
..			File folder		
botgame.c	1.240	672	C Source	29/09/2017 11:...	4823F73B
ex0.c	1.349	514	C Source	29/09/2017 11:...	7B750777
ex0a.c	633	355	C Source	29/09/2017 11:...	7D4A75CB
ex0b.c	1.187	554	C Source	29/09/2017 11:...	FA6E7A36
makefile	557	299	File	29/09/2017 11:...	66B8D18E
oppgaver.txt	2.251	1.148	Text Document	29/09/2017 11:...	A2E981C2

There are a total of 4 tasks in this exercise, see the rest of this presentation for more details on some of them.

Practical

```
wget http://www.eastwill.no/pg3401/exercises_05.zip
```

```
unzip exercises_05.zip
```

```
ls
```

```
cd Leksjon5
```

```
cat oppgaver.txt
```

- Follow the instructions

You can also download this from Canvas.

Dynamic multidim array (task 1)

Creating an array

First part of the task is simple;

- Find the size of the array (ask the user)
- Ask user for a value
- Select row at random

To select a number at random use the `rand()` function. Recommend reading about that function (use `man`), but the short version is that you should first call `srand(time(NULL))` ;

Printing an array

- Experiment with printing using `\t` to get a tabulator
- For advanced students you could also try to use special characters such as `_` and `⌋` and `└` and `┐` to “draw” a UI using printable characters

Bubble sorting

- We will not require much use of “algorithms and data structures” in this course, but simple things are expected
 - Bubble sorting (last weeks exercises)
 - Linked lists (more in the coming weeks)

String length (task 2)

String length

- This is a very simple task, I hope everyone can manage this in a few minutes 😊

String traversal (task 3)

String traversal

- Try to accomplish this task in 2 ways:
 - An integer traversing the string by index
 - A pointer traversing the string

A game (task 4)

This is a difficult task that might not be suitable for all students, and might take a lot of time for some

Coding an enemy bot

Menacing enemy for sure :-)

- This is a difficult task that might not be suitable for all students, and might take a lot of time for some
- Understanding the concepts of the game is the first step, then starting build the code by inspecting botgame.c for some inspiration and determinate structures

010

101

110

*You need some random numbers, so use
`rand()` and `srand(time(NULL))` ;*

If this task is too difficult

- If this task is too difficult for you then focus on completing all tasks in the previous exercises
- In Canvas I have also uploaded the 2021 exam, you can start on task 2 this week – we have not covered reading/writing files yet, but you can hardcode the input text-string as a char array and solve the rest of the task

Deloppgave A

Skriv et program som leser denne filen, dekode dataene, og skriver resultatet til en annen fil.

Både input filen og output filen skal være i samme katalog som programmet, og begge filene skal være en del av innlevert besvarelse.

Eksempel: Første «tegn» i datasettet er 44, dette er med andre ord ASCII koden 0x44, som er en stor bokstav D. I output filen skal du derfor skrive D for dette første «tegnet».

Deloppgave B

Du skal så lage en funksjon som tar de dekodete dataene som parameter og teller opp antall forekomster av hver bokstav (A til Z), funksjonen skal telle både store og små bokstaver som samme bokstav, og skal overse andre tegn. Programmet ditt skal skrive ut i konsoll vinduet antall forekomster av hver bokstav.

Prepare for next week

Tidligere_PG3401-Hjemmeeksamen-14dager-H21.pdf

Prepare for next week

- In Canvas I have uploaded the 2021 exam
- Next week we have a look at task 3 of that exam (the exam text is in Norwegian only, use google to translate)
- Recommend that you have a look to prepare :-)

Du skal lage funksjoner som utfører følgende operasjoner på listen:

- Legge til et element i slutten av listen
- Slette siste element i listen (en «angreknapp»)
- Sletter alle elementer i listen som har et gitt VARENAVN (sendes som parameter til funksjonen)
- Summerer sammen total pris for varene i listen (husk å ta hensyn til antall)
- Printer ut en kvittering (i konsoll vinduet)



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